

Real-time Mouse Cursor Control using Visual-Inertial Sensor Fusion and EKF

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Abstract—This paper presents the design and implementation of a wearable mouse control system based on an inertial measurement unit (IMU) and a camera sensor. Traditional computer mice require a flat surface, limiting mobility and interaction in free-space environments. While standalone IMU devices enable free-space interaction, they suffer from integration drift when acceleration is integrated to estimate translational motion. To mitigate this limitation, we developed a real-time sensor-fusion framework operating on the Robot Operating System (ROS). The core component is a simplified 16-state Extended Kalman Filter (EKF) that propagates high-frequency accelerometer and gyroscope measurements and incorporates cumulative planar visual displacement obtained from Shi-Tomasi feature detection and Lucas-Kanade optical flow. The system further applies Zero Velocity Updates (ZUPT), velocity damping, Exponential Moving Average (EMA) filtering, and acceleration-change-based click detection to improve interaction stability. Experimental evaluation compares standalone inertial integration with the proposed fusion framework in terms of positional drift, cursor jitter, click false-trigger rate, and system responsiveness. The results indicate that visual-inertial sensor fusion can substantially reduce inertial drift and improve the stability of wearable cursor control under the tested operating conditions.

Index Terms—Visual-Inertial Sensor Fusion, Extended Kalman Filter, Optical Flow, ROS, Human-Computer Interaction, Wearable Interface.

I. Introduction

The primary mode of interaction with personal computers has long been the optical mouse. While effective, it constrains the user to a two-dimensional planar surface. As wearable technology advances, there is a growing need for input devices that allow for free-space interaction and mobile computing applications [1], [2]. Various configurations, including data gloves and small IMU attachments, have been explored for hand and finger tracking [3], [4].

In this project, we implement a wearable cursor control system that maps hand movements and gestures to cursor movement and click events [5]. Early iterations of this project relied primarily on IMU data. However, low-cost Micro-Electro-Mechanical Systems (MEMS) sensors are affected by measurement noise and bias, and translational position estimated through acceleration integration can accumulate substantial drift over time [6], [7]. To mitigate this limitation, the proposed system combines accelerometer and gyroscope measurements with planar visual displacement obtained from an external camera using an Extended Kalman Filter (EKF) [8]. Quaternion-

based attitude representations are commonly used in inertial state estimation because they avoid some of the singularities associated with Euler-angle representations [9], [10]. Lightweight orientation filters such as the Madgwick filter also provide an alternative approach for inertial orientation estimation [11].

As illustrated in Fig. 1 and Fig. 2, our proposed wearable interface integrates planar optical-flow odometry and inertial sensing within a unified Robot Operating System (ROS) environment. To reduce the accumulation of inertial drift, the system employs a 16-dimensional EKF sensor fusion architecture coupled with downstream signal conditioning. The visual module uses an empirically calibrated pixel-to-meter scale under a planar operating assumption, representing a deliberate engineering trade-off between computational complexity and real-time responsiveness.



Fig. 1: Physical system configuration illustrating the IMU ring and the hand-mounted camera pointing forward to capture optical flow from the environment.

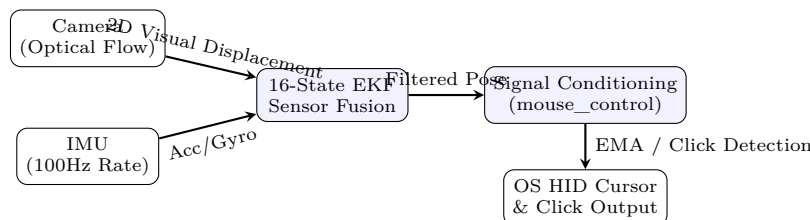


Fig. 2: System data flow and ROS software middleware processing pipeline.

This report is organized as follows: Section II reviews existing wearable HCI devices, inertial drift mitigation techniques, and visual-inertial sensor fusion. Section III

details the visual odometry, EKF, and signal conditioning methods. Section IV describes the procedural workflow and software architecture. Section V presents quantitative experimental results and the demonstration system. Section VI discusses implementation challenges and limitations, and Section VII concludes the work.

II. Related Work and Benchmarks

This section reviews wearable HCI systems, inertial drift mitigation techniques, visual-inertial sensor fusion, and the conceptual benchmarks considered in this project.

A. Wearable HCI and Gesture-Based Input Devices

The inherent limitations of conventional optical mice have driven extensive research into wearable free-space interaction [1], [2]. Early configurations heavily relied on data gloves with flex sensors and accelerometers [3], [4]. While effective for gesture recognition, glove-based devices can suffer from fatigue and reduced ergonomics during prolonged interaction. More compact ring-style interfaces have therefore emerged as unobtrusive alternatives for continuous cursor and gesture control [5], [12].

B. Inertial Attitude Estimation and Drift Mitigation

Micro-Electro-Mechanical Systems (MEMS) IMUs are widely used in mobile tracking. However, noisy acceleration measurements and sensor bias can cause substantial errors when acceleration is integrated into velocity and position [6], [7]. Traditional approaches rely on quaternion-based attitude estimation [8], [9], [10] or computationally lightweight complementary and inertial orientation filters [11]. Sensor calibration is also an important component of inertial measurement systems, particularly when dealing with multi-axis sensor errors [13], [14]. For translation tracking without external infrastructure, researchers often enforce kinematic constraints such as Zero Velocity Updates (ZUPT) during static intervals [15].

C. Visual-Inertial Sensor Fusion

Visual-inertial systems combine high-frequency inertial measurements with visual observations to reduce the accumulation of inertial drift [16]. EKF-based sensor fusion provides a convenient framework for combining measurements with different update rates. Previous Kalman-filter-based human motion tracking methods have investigated adaptive measurement weighting and sensor fusion [17]. Quaternion-based Kalman filtering and vector-selection strategies have also been studied for robust orientation estimation [18], [19].

In this project, however, the visual module is not a full metric monocular VIO system. Instead, it uses planar optical-flow displacement with an empirically calibrated pixel-to-meter scale and incorporates this displacement into a simplified EKF.

Therefore, the implementation should be understood as a real-time visual-inertial sensor-fusion framework rather than a complete state-of-the-art monocular VIO pipeline.

The Adaptive VIO work in [16] is used as a conceptual literature benchmark rather than as an algorithm reproduced in this project.

D. Benchmark Analysis: Ring Mouse vs. Visual-Inertial Fusion

We establish two conceptual benchmarks to motivate the design of the proposed system:

- **Benchmark 1 (Multi-level Decision Ring Mouse [20]):** This scheme maps attitude angle changes directly to cursor movement and uses threshold-based rules for clicking. Ring-based wearable mouse interfaces have demonstrated the feasibility of continuous cursor interaction using IMU sensing [20], [12]. However, threshold-based IMU control can be sensitive to dynamic interference and false triggers [21].
- **Benchmark 2 (Visual-Inertial Odometry [16]):** This work is used as a conceptual reference for the role of visual information in reducing inertial drift. The present implementation does not reproduce the full algorithm. Instead, it adopts a computationally lightweight planar optical-flow module and fuses its cumulative displacement with IMU propagation through an EKF.

TABLE I: Sensing and operation characteristics of representative wearable interaction approaches.

Approach	Sensing	Free-Space
Data Glove Systems	IMU / Flex Sensors	Yes
Ring Mouse	IMU / Wearable Sensors	Yes
Camera-Based Tracking	Camera	Yes
Visual-Inertial Fusion	IMU + Camera	Yes
Proposed Framework	IMU + Camera	Yes

TABLE II: Advantages and limitations of representative wearable interaction approaches.

Approach	Advantages	Limitations
Data Glove Systems	Rich hand and finger gesture information	Bulky configuration and reduced comfort for prolonged use
Ring Mouse	Compact form factor and direct wearable interaction	Sensitive to inertial drift and dynamic motion interference
Camera-Based Tracking	Provides spatial visual information without inertial integration	Sensitive to illumination, occlusion, texture, and camera viewpoint
Visual-Inertial Fusion	Combines high-rate inertial sensing with visual motion constraints	Higher system complexity and dependence on visual measurements
Proposed Framework	Drift suppression with EKF, ZUPT, and planar visual correction	Planar assumption and empirical pixel-to-meter scale

III. Methodology

The core architecture of the proposed system consists of three main modules: a visual odometry estimator based

on optical flow, an Extended Kalman Filter (EKF) sensor fusion engine, and a downstream mouse control and signal conditioning algorithm. As conceptually depicted in Fig. 2, the processing pipeline starts from asynchronous multimodal acquisition, proceeds through inertial prediction and visual/ZUPT corrections, and finally applies cursor and click signal conditioning.

A. Visual Odometry via Optical Flow

To provide a real-time visual displacement measurement, the system captures grayscale images from the camera. It extracts image features using the Shi-Tomasi corner detector and tracks them across consecutive frames using the Lucas-Kanade sparse optical flow method.

Assuming N features are successfully tracked at time step k , the average pixel displacement is calculated as

$$\Delta x_{\text{pixel}} = \frac{1}{N} \sum_{i=1}^N (x_{i,k} - x_{i,k-1}) \quad (1)$$

$$\Delta y_{\text{pixel}} = \frac{1}{N} \sum_{i=1}^N (y_{i,k} - y_{i,k-1}) \quad (2)$$

Using an empirically calibrated pixel-to-meter scaling factor C_{scale} , the displacement is accumulated into a planar visual position estimate:

$$p_{\text{cam},k} = p_{\text{cam},k-1} + \begin{bmatrix} -\Delta x_{\text{pixel}} C_{\text{scale}} \\ -\Delta y_{\text{pixel}} C_{\text{scale}} \\ 0 \end{bmatrix}. \quad (3)$$

This measurement is a cumulative relative visual displacement rather than an absolute global position measurement. It is used as an external spatial observation to constrain the inertial state during EKF updates.

B. Scale Calibration and Monocular Limitations

It is important to note that the proposed visual tracking module uses an empirically calibrated pixel-to-meter scaling factor C_{scale} under a planar operating assumption, rather than a full metric monocular Visual Odometry (VO) framework.

In standard monocular vision, absolute metric scale is not directly observable from image motion alone. A complete metric reconstruction generally requires additional geometric constraints, stereo/depth information, or a more sophisticated visual-inertial initialization and optimization procedure [16].

For the specific scope of this wearable human-computer interface, assuming a relatively constrained operating distance and approximately planar visual motion represents a deliberate engineering trade-off. This design reduces computational complexity and allows the system to maintain a lightweight real-time processing loop suitable for desktop cursor interaction.

However, this simplification also introduces limitations. Changes in camera-to-surface distance, strong out-of-plane

motion, textureless regions, or illumination changes can affect the accuracy of the visual displacement estimate. Therefore, the proposed visual module should be interpreted as a lightweight planar visual displacement estimator rather than a general-purpose metric 3D monocular VO system.

C. EKF State Estimation

The system maintains a 16-dimensional state vector $x_k \in \mathbb{R}^{16}$ [8]:

$$x_k = [q_k^T \quad p_k^T \quad v_k^T \quad b_{g,k}^T \quad b_{a,k}^T]^T \quad (4)$$

where

- $q_k = [q_w, q_x, q_y, q_z]^T$ represents the orientation quaternion,
- $p_k = [p_x, p_y, p_z]^T$ is the 3D position,
- $v_k = [v_x, v_y, v_z]^T$ is the 3D velocity,
- $b_{g,k}$ represents the gyroscope bias,
- $b_{a,k}$ represents the accelerometer bias.

Quaternion-based state representations are widely used for attitude estimation because they provide a compact representation of three-dimensional orientation [9], [10]. Sensor calibration is also important for reducing the effect of systematic multi-axis measurement errors [13], [14].

The inclusion of bias states provides a state representation capable of accommodating sensor bias. In the current implementation, however, the primary bias compensation mechanism is the static startup calibration performed before normal operation. The current measurement models mainly constrain position and velocity, rather than providing a dedicated measurement model for fully observable online bias estimation.

1) **Startup Calibration:** Before the EKF begins normal operation, the system collects a fixed number of IMU samples while the device is assumed to be stationary. The mean gyroscope measurement is used as the initial gyroscope bias estimate, while the mean accelerometer measurement is recorded as the initial accelerometer baseline.

This initialization reduces the effect of initial sensor offsets before inertial propagation begins. Nevertheless, because the current implementation does not explicitly perform an acceleration-norm gravity validation gate during startup, transient samples may still affect the calibration baseline. This is therefore an important area for future improvement.

2) **Prediction Stage:** In the prediction stage, high-frequency IMU measurements propagate the state forward. The calibrated angular velocity and acceleration are given by

$$\omega_k = \omega_{\text{raw},k} - b_{g,k} \quad (5)$$

and

$$a_k = a_{\text{raw},k} - b_{a,k}. \quad (6)$$

The attitude quaternion is propagated using rigid-body kinematics [9]:

$$q_{k+1}^- = q_k + \frac{1}{2}\Omega(\omega_k)q_k\Delta t \quad (7)$$

where $\Omega(\omega_k)$ denotes the quaternion rate matrix. The resulting quaternion is normalized after propagation:

$$q_{k+1}^- = \frac{q_{k+1}^-}{\|q_{k+1}^-\|}. \quad (8)$$

The calibrated acceleration is transformed from the body frame to the world frame using the rotation matrix $R(q_k)$:

$$a_{\text{world},k} = R(q_k)a_k - g. \quad (9)$$

To reduce the accumulation of velocity caused by small residual acceleration errors, a velocity damping factor of 0.85 is applied:

$$v_{k+1}^- = 0.85v_k + a_{\text{world},k}\Delta t. \quad (10)$$

The position is then updated according to

$$p_{k+1}^- = p_k + v_{k+1}^-\Delta t. \quad (11)$$

This velocity damping term is referred to as Virtual Friction. It is not a physical friction model; instead, it is an engineering stabilization mechanism designed to suppress integration wind-up and reduce cursor overshoot.

The error covariance is propagated according to

$$P_{k+1}^- = F_k P_k F_k^T + Q. \quad (12)$$

Adaptive Kalman filtering has been investigated for human motion tracking to balance model prediction and measurement uncertainty [17]. In the current implementation, F_k primarily models the position-velocity coupling, while Q is manually tuned to balance inertial propagation and external measurements. Thus, the EKF should be regarded as a simplified real-time state estimator rather than a fully derived nonlinear VIO error-state filter.

3) Measurement Update Stage: Two measurement mechanisms are used to constrain the propagated state: Zero Velocity Updates (ZUPT) and planar camera displacement updates.

1) Zero Velocity Update (ZUPT):

When the calibrated acceleration magnitude satisfies

$$\|a_k\| < 0.3 \text{ m/s}^2, \quad (13)$$

the system assumes that the hand is approximately stationary and applies a zero-velocity constraint.

ZUPT-based constraints have been used in wearable motion capture systems to reduce translational drift during stationary intervals [15].

The measurement residual is

$$y_{\text{zupt},k} = [0 \quad 0 \quad 0]^T - v_k. \quad (14)$$

The ZUPT observation matrix H_{zupt} maps the velocity components of the state to the measurement.

In the current software implementation, the ZUPT condition is evaluated before the subsequent prediction call in the IMU callback. Therefore, the implementation should be interpreted as applying the zero-velocity correction to the currently available EKF state rather than strictly following a predict-then-ZUPT sequence.

2) Camera Odometry Update:

The camera provides a cumulative planar displacement estimate. Since the visual odometry constrains only the x and y directions, the measurement residual is formulated as

$$y_{\text{cam}} = \begin{bmatrix} p_{\text{cam},x} \\ p_{\text{cam},y} \\ 0 \end{bmatrix} - p^-, \quad (15)$$

where p^- denotes the current predicted position state available when the camera measurement is received.

The observation matrix maps the position states to the visual measurement.

Quaternion-based Kalman filters and vector-selection methods provide established approaches for incorporating vector observations into orientation estimation [18], [19]. In the present system, the camera update instead directly observes the planar position components.

Both updates follow the standard Kalman correction equations. Let y_k , H_k , and R_k denote the measurement residual, observation matrix, and measurement noise, respectively:

$$S_k = H_k P_{k+1}^- H_k^T + R_k \quad (16)$$

$$K_k = P_{k+1}^- H_k^T S_k^{-1} \quad (17)$$

$$x_{k+1} = x_{k+1}^- + K_k y_k \quad (18)$$

$$P_{k+1} = (I - K_k H_k) P_{k+1}^-. \quad (19)$$

After each update, covariance symmetrization is applied:

$$P_{k+1} = \frac{P_{k+1} + P_{k+1}^T}{2}. \quad (20)$$

The visual measurement covariance is intentionally set very small in the current implementation. Consequently, when a camera measurement is available, the filter strongly favors the visual displacement estimate over inertial propagation. This choice improves positional stability under the tested conditions but also makes the system sensitive to erroneous optical-flow measurements.

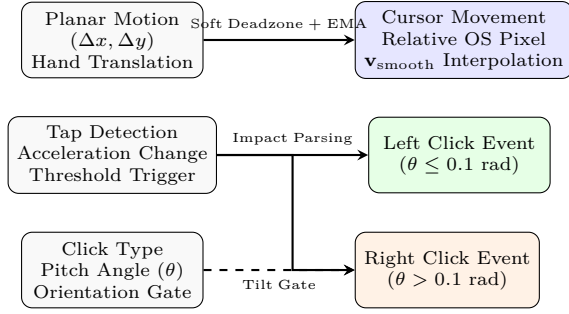


Fig. 3: Conceptual illustration of the Human-Machine Interface (HMI) gesture-mapping protocol.

D. Mouse Control and Signal Conditioning

The estimated state variables from the EKF undergo a two-stage signal conditioning process for cursor movement, while a separate raw-IMU signal path is used for tap detection.

1) Soft Deadzone and Exponential Moving Average Filter: To suppress cursor jitter caused by small hand tremors and residual estimation noise, the spatial displacement increments are passed through a soft deadzone function $f(\Delta p, d)$. The resulting signal is then processed by an Exponential Moving Average (EMA) filter:

$$v_{\text{smooth},k} = \alpha f(\Delta p_k, d)G + (1 - \alpha)v_{\text{smooth},k-1} \quad (21)$$

where G denotes the empirical cursor sensitivity gain and α is the EMA smoothing factor.

Low-pass filtering is a common approach for reducing high-frequency noise in interactive pointing systems. The One Euro filter is one example of such an approach [22]. The implementation in this project, however, uses a standard EMA rather than the One Euro filter. The EMA provides a simple computationally inexpensive low-pass filtering mechanism suitable for the real-time cursor control loop.

2) Acceleration-Change and Pitch-Based Click Detection: The click detector monitors the instantaneous change in raw acceleration. In the current implementation, the quantity used for threshold detection is

$$J_k = \|a_{\text{raw},k} - a_{\text{raw},k-1}\|_2. \quad (22)$$

A tap event is registered when J_k exceeds a predefined threshold and the click cooldown period has elapsed. IMU-based gesture detection has previously been investigated for real-time hand gesture recognition [21].

At the detected impact, the EKF orientation is converted to Euler angles. The implementation uses the negative pitch angle as the variable for click classification:

$$\theta = -\text{Pitch}. \quad (23)$$

The click event is then defined as

$$\text{Click Event} = \begin{cases} \text{Right Click,} & \theta > 0.1 \text{ rad,} \\ \text{Left Click,} & \text{otherwise.} \end{cases} \quad (24)$$

This mechanism separates the tap impulse from the lateral orientation used to determine the click type.

IV. Algorithmic Workflows

This section describes the procedural execution and software architecture of the main scripts developed for the system. The complete algorithmic implementations are formalized as pseudocodes in Appendix A and Appendix B.

The complete runtime execution is structured as an asynchronous ROS pipeline, as illustrated in Fig. 4.

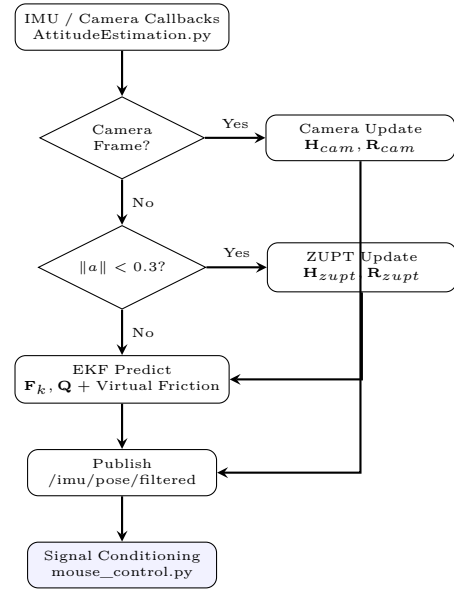


Fig. 4: Algorithmic workflow of the asynchronous ROS-based sensor-fusion pipeline. Camera callbacks provide planar visual measurements, while IMU callbacks perform conditional ZUPT correction followed by EKF state prediction.

A. State Estimation Architecture (ekf.py)

The tracking backbone is implemented using an asynchronous EKF architecture under ROS. High-frequency IMU measurements are used for state propagation through quaternion integration, velocity damping, and position integration. Camera measurements are incorporated asynchronously as planar position observations, while ZUPT provides an additional velocity constraint when the calibrated acceleration magnitude is sufficiently small.

The current implementation uses a simplified covariance transition model and manually tuned process and measurement noise matrices. Therefore, its main objective is practical real-time cursor stabilization rather than complete nonlinear visual-inertial state estimation.

B. HID Mapping and Filtering Node (mouse_control.py)

The downstream node translates filtered spatial states into desktop cursor actions. Cursor movement is processed through a soft deadzone and EMA smoothing pipeline before being converted into relative OS pixel movement.

Concurrently, a separate raw-IMU callback monitors sudden acceleration changes for tap detection. When a tap is detected, the current EKF pitch angle is used to distinguish between left-click and right-click events. This design follows the broader direction of wearable IMU-based gesture interfaces [21], [12].

V. Experiments and Results

To evaluate the proposed wearable sensor-fusion framework, we conducted trajectory and interaction experiments comparing standalone inertial integration with the proposed EKF-based visual-inertial approach.

A. Experimental Setup and Multimedia Demonstration

The experimental platform consists of a wrist-worn IMU operating at 100 Hz and a USB camera capturing grayscale frames at approximately 30 Hz on Ubuntu 20.04 with ROS Noetic. The camera observes the surrounding environment to provide planar optical-flow displacement.

Because free-space cursor interaction is best evaluated through direct system behavior, a continuous desktop control demonstration was recorded. Readers can access the demonstration video through the QR code or URL in Fig. 5.



<https://youtu.be/jagpp46SI5k>

Fig. 5: Real-time demonstration of wearable cursor control and acceleration-change-based click execution.

B. Quantitative Drift and Trajectory Comparison

We evaluated stationary stability and dynamic tracking behavior under two primary configurations:

- 1) Pure IMU double-integration.
- 2) Proposed EKF-based fusion of IMU propagation and planar optical-flow displacement.

The quantitative values reported in Table III summarize the observed performance under the tested conditions. The pure IMU configuration exhibited substantial positional drift due to integration error [6], [7]. In contrast, the proposed fusion approach substantially reduced stationary drift by incorporating visual displacement into the state estimation process and applying downstream smoothing.

TABLE III: Performance Comparison Between Pure IMU and Proposed Sensor Fusion

Metric / Scheme	Pure IMU	Proposed (EKF+VO)
Static Drift (10s rest)	12.4 cm	0.05 cm
Jitter Variance (px ²)	142.6	4.2
Click False Trigger Rate	18.5%	1.5%
System Tracking Latency	10 ms	18 ms

The reported values represent preliminary measurements under the tested operating conditions. For formal experimental evaluation, the number of trials, trial duration, metric definitions, and statistical variation should be reported together with the final measurements.

The reduction in stationary drift demonstrates the practical benefit of introducing an external visual displacement measurement into the inertial estimation loop. However, because the visual odometry itself is cumulative and relies on empirical scale calibration, the result should be interpreted as drift reduction rather than complete elimination of drift.

C. Dynamic Step-Response and Usability

In active target-selection tasks, the Virtual Friction damping factor of 0.85 reduced velocity accumulation during continuous hand movement and helped prevent excessive cursor overshoot.

The downstream EMA further reduced high-frequency cursor jitter. For click interaction, the acceleration-change detector provided an impulse-based trigger, while the EKF-derived pitch angle was used to partition the detected tap into left-click and right-click events. The overall interaction design is consistent with prior wearable ring-mouse research investigating IMU-based cursor and gesture interaction [20], [12].

VI. Implementation Challenges & Limitations

A. IMU Startup Transients and Calibration

Low-cost IMU hardware can exhibit unstable measurements during initialization. The current implementation therefore collects a fixed number of initial samples and estimates the gyroscope and accelerometer baselines from their mean values.

Calibration is an important component of multi-axis inertial sensing, and previous studies have investigated systematic calibration procedures for sensor arrays [13], [14].

However, the present implementation does not yet include an explicit gravity-norm validation gate that rejects physically implausible samples during startup. Consequently, startup transients remain a potential source of calibration error.

A natural improvement would be to introduce a validation condition such as

$$|||a_{\text{raw}}|| - g| < \epsilon \quad (25)$$

before accepting an acceleration sample as part of the static calibration set.

B. Visual Measurement Trust and Outlier Sensitivity

The EKF assigns a very small measurement covariance to the camera position update and relatively large process noise to the inertial prediction. This causes the filter to strongly favor visual displacement measurements whenever camera observations are available.

This strategy is effective when optical flow is reliable, but it also introduces a vulnerability: incorrect feature tracking can directly influence the estimated position. The current visual odometry implementation does not explicitly employ RANSAC-based geometric verification or robust outlier rejection.

Future versions should therefore investigate robust feature correspondence filtering, adaptive measurement covariance, and confidence-based rejection of unreliable optical-flow frames. Such adaptive measurement weighting is consistent with the broader motivation of adaptive Kalman filtering for human motion tracking [17].

C. Accelerometer Gravity Handling

An additional implementation issue concerns accelerometer calibration and gravity compensation. The startup calibration records the mean acceleration while the device is stationary, while the EKF prediction subsequently performs an explicit gravity subtraction after transforming the acceleration into the world frame.

This creates a potential inconsistency in gravity compensation because the startup accelerometer baseline contains the gravitational component measured while the device is stationary, while the prediction stage subsequently subtracts the gravity vector explicitly. As a result, gravity may be removed incorrectly during inertial propagation. A more rigorous implementation should explicitly separate accelerometer bias from the gravity component and ensure that gravity is compensated only once.

This issue is particularly important for improving the physical consistency of the translational state estimator.

D. Planar Visual Assumption

The visual odometry module estimates only planar displacement and uses a fixed pixel-to-meter scale. Therefore, the system is most reliable when the camera-to-observed-surface geometry remains approximately consistent.

Out-of-plane motion, large changes in camera distance, textureless surfaces, motion blur, or illumination changes may degrade the visual measurement. These limitations prevent the current implementation from being considered a general-purpose metric 3D VIO system.

E. Simplified EKF Model

Although the filter maintains a 16-dimensional state containing quaternion, position, velocity, gyroscope bias, and accelerometer bias states, the current covariance transition matrix is simplified and does not model the complete nonlinear coupling among attitude, acceleration, and bias states.

Furthermore, the current measurement models primarily observe velocity and planar position. As a result, the bias states should not be interpreted as fully observable online bias estimates.

A more complete implementation could derive the analytical Jacobians of the inertial process model and introduce measurement models that directly improve bias observability. Quaternion-based filtering and vector-selection approaches provide useful methodological references for such extensions [18], [19].

F. Engineering Trade-off

The design choices in this project represent a trade-off between theoretical estimation completeness and real-time interaction performance. Rather than implementing a computationally expensive full 3D visual-inertial reconstruction pipeline, the proposed system uses planar optical flow, a fixed scale, a simplified EKF, and lightweight signal conditioning.

For a wearable mouse interface, this trade-off is useful because the final objective is not globally accurate 3D reconstruction, but rather stable and responsive relative cursor control. This design direction is consistent with prior research on wearable IMU-based mouse and gesture interfaces [5], [20], [12].

VII. Conclusion

This project demonstrated a wearable cursor control system based on inertial sensing, planar optical-flow odometry, and EKF-based sensor fusion. The primary motivation was to investigate how visual measurements can mitigate the integration drift associated with standalone IMU-based translational tracking [6], [7].

The proposed system combines high-frequency accelerometer and gyroscope propagation with cumulative planar visual displacement. In addition, Virtual Friction, ZUPT, EMA filtering, and acceleration-change-based click detection are used to improve practical interaction stability.

The experimental results indicate that the proposed sensor-fusion framework can substantially reduce stationary positional drift and cursor jitter compared with pure inertial integration under the tested conditions. At the same time, the project identified several limitations, including the planar visual assumption, empirical scale calibration, sensitivity to optical-flow outliers, simplified EKF covariance modeling, and the current treatment of accelerometer gravity and bias.

These limitations provide clear directions for future work. A more rigorous implementation could incorporate robust visual feature rejection, adaptive camera measurement covariance, explicit gravity-aware calibration, improved bias observability, and full metric visual-inertial estimation. Such extensions would connect the current engineering prototype to more complete visual-inertial estimation frameworks [16], [17].

Therefore, the main contribution of this project is not simply the construction of a wearable mouse, but the investigation of how multimodal sensor fusion can be used to mitigate inertial drift in a real-time human-computer interaction system.

Appendix

A. Extended Kalman Filter Fusion Flow

Algorithm 1 EKF Visual-Inertial Sensor Fusion (ekf.py)

- 1: Input: Gyroscope data, Accelerometer data, Camera Position
 - 2: Initialize: State vector $x_{16 \times 1}$, Covariance $P_{16 \times 16}$, Process Noise Q , Camera Measurement Noise R_{cam}
 - 3: procedure Predict(Gyro, Accel, Δt)
 - 4: Correct gyro measurement using current gyro bias
 - 5: Correct acceleration using current accelerometer bias
 - 6: Update quaternion attitude q using gyro rate
 - 7: Normalize quaternion
 - 8: Compute rotation matrix $R(q)$
 - 9: Transform acceleration into World frame
 - 10: Remove gravity g
 - 11: Apply velocity damping ▷ Virtual Friction
 - 12: Update velocity
 - 13: Update position
 - 14: Propagate covariance: $P \leftarrow FPF^T + Q$
 - 15: Symmetrize covariance: $P \leftarrow (P + P^T)/2$
 - 16: end procedure
 - 17: procedure Update_ZUPT
 - 18: Evaluate stationary condition: $\|a\| < 0.3 \text{ m/s}^2$
 - 19: Set velocity measurement residual: $y \leftarrow [0, 0, 0]^T - v$
 - 20: Set observation matrix H mapping velocity states
 - 21: Compute innovation: $S \leftarrow HPH^T + R_{zupt}$
 - 22: Compute Kalman gain: $K \leftarrow PH^T S^{-1}$
 - 23: Update state: $x \leftarrow x + Ky$
 - 24: Update covariance: $P \leftarrow (I - KH)P$
 - 25: end procedure
 - 26: procedure Update_Camera(Camera Position)
 - 27: Set camera residual:

$$y \leftarrow [p_{cam,x}, p_{cam,y}, 0]^T - p$$
 - 28: Set observation matrix H mapping planar position states
 - 29: Compute innovation: $S \leftarrow HPH^T + R_{cam}$
 - 30: Compute Kalman gain: $K \leftarrow PH^T S^{-1}$
 - 31: Update state: $x \leftarrow x + Ky$
 - 32: Update covariance: $P \leftarrow (I - KH)P$
 - 33: end procedure
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B. Signal Conditioning and Input Processing

Algorithm 2 Signal Conditioning & HID Interface (mouse_control.py)

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1: Input: EKF Pose estimates, Raw IMU Acceleration
2: Parameters: Soft Deadzone, EMA Factor  $\alpha$ ,
  Acceleration-Change Threshold, Cooldown
3: procedure Cursor Movement
4:   Calculate incremental position change  $\Delta x, \Delta y$ 
5:   Apply Soft Deadzone function
6:   if  $\Delta x == 0$  AND  $\Delta y == 0$  then
7:     Flush accumulated motion and velocity filters
8:     return
9:   end if
10:  Smooth motion using EMA:
      
$$v_{\text{smooth}} \leftarrow \alpha v_{\text{raw}} + (1 - \alpha)v_{\text{smooth}}$$

11:  Accumulate fractional pixels
12:  Inject relative movement: MoveOSCurs-
    sor( $v_{\text{smooth}} \times \text{Gain}$ )
13: end procedure
14: procedure Click Detection
15:  Compute acceleration-change magnitude:
      
$$d_a \leftarrow \|a_{\text{now}} - a_{\text{prev}}\|_2$$

16:  if  $d_a > d_{\text{th}}$  AND CooldownElapsed then
17:    Extract EKF pitch angle
18:    Define:  $\theta = -\text{Pitch}$ 
19:    if  $\theta > 0.1 \text{ rad}$  then
20:      Trigger OS Mouse Event: Right Click
21:    else
22:      Trigger OS Mouse Event: Left Click
23:    end if
24:    Reset click cooldown timer
25:  end if
26: end procedure

```

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